



Integrating artificial intelligence and Internet of Things (IoT) for enhanced crop monitoring and management in precision agriculture

Kushagra Sharma^a, Shiv Kumar Shivandu^{b,*}

^a Department of Horticulture (Fruit Science), Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur, Madhya Pradesh, 482004, India

^b Department of Horticulture (Fruit Science), College of Horticulture, Dr YS Parmar University of Horticulture and Forestry, Nauni, Solan, Himachal Pradesh, 173230, India

ARTICLE INFO

Keywords:

Precision agriculture
Artificial Intelligence (AI)
Internet of Things (IoT)
Crop monitoring
Smart farming

ABSTRACT

The integration of Artificial Intelligence (AI) and Internet of Things (IoT) technologies is transforming precision agriculture by enhancing crop monitoring and management. This review explores cutting-edge methodologies and innovations in modern agriculture, including high-throughput phenotyping, remote sensing, and automated agricultural robots (AgroBots). These technologies automate tasks such as harvesting, sorting, and weed detection, significantly reducing labor costs and environmental impacts. High-throughput phenotyping leverages remote sensing, spectral imaging, and robotics to collect data on plant traits, enabling informed decisions on fertilization, irrigation, and pest management. DGPS and remote sensing offer precise, real-time data essential for soil condition assessment and crop health monitoring. Advanced image segmentation techniques ensure accurate detection of plants and fruits, overcoming challenges posed by varying lighting conditions and complex backgrounds. Case studies like the PACMAN SCRI project for apple crop load management and Project PANTHEON's SCADA system for hazelnut orchard management demonstrate the transformative potential of AI and IoT in optimizing agricultural practices. The upcoming integration of 5G and future 6G mobile networks promises to address connectivity challenges, promoting the widespread adoption of smart agricultural practices. However, several research gaps remain. Integrating diverse datasets, ensuring scalability for small and medium-sized farms, and enhancing real-time decision-making need further investigation. Developing robust AI models and IoT devices for varied agricultural conditions, creating user-friendly interfaces for farmers, and addressing privacy and security concerns are essential. Addressing these gaps can enhance the effectiveness and adoption of AI and IoT in precision agriculture, leading to more sustainable and productive farming practices.

1. Introduction

Precision agriculture entails utilizing information and communication technology (ICT) to manage the spatial and temporal variability of fields. By assessing these variations in field and crop characteristics, areas with similar traits, known as management zones, can be identified. Professor Pierre C. Robert, is often considered pioneer of precision farming, described it as an information revolution facilitated by new technologies. He highlighted that precision agriculture transcends merely adopting new technologies; it establishes a more precise and sophisticated farm management system [1]. In the years ahead, food production needs to rise substantially to satisfy the demands of a global population projected to reach 9.7 billion by mid-century [2]. To address field variability, a range of technologies has been developed, utilizing

ground-based sensing systems. These systems include mobile platforms and tractors equipped with cameras, ultrasonic sensors (sonar), and light detection and ranging (LiDAR) sensors, also known as laser sensors.

The large-scale mechanization of agriculture in the twentieth century replaced labor with machinery, boosting land productivity and achieving economies of scale. Precision farming holds significant promise for increasing farmer incomes, enhancing both the extrinsic and intrinsic quality of agricultural products, and reducing the adverse environmental effects of farming [3]. Although not a cure-all, precision farming can significantly contribute to more sustainable agricultural practices. The ongoing fourth industrial revolution is revolutionizing agriculture, emerging in the era of Agriculture 4.0. This new phase is defined by data-driven management, innovative tool-based production, sustainability, professionalization, and a minimized environmental

* Corresponding author.

E-mail address: marginshiv05@gmail.com (S.K. Shivandu).

<https://doi.org/10.1016/j.sintl.2024.100292>

Received 1 July 2024; Accepted 3 August 2024

Available online 3 August 2024

2666-3511/© 2024 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

impact [4]. Key components of Agriculture 4.0 include modern smart technologies such as robotics (including drones), big data, artificial intelligence, computer vision, 5G, cloud computing, the Internet of Things, and blockchain technology. The venin diagram presented in Fig. 1 shows fundamentals on which the precision farming works along with the agricultural tools and techniques which are currently useful for precision agriculture.

Agriculture accounts for 85 % of the total water managed globally, underscoring the necessity for precise water management in orchards, particularly in semi-arid regions, in these areas water input is substantial economic investment. Climate change further complicates these challenges for the fruit industry by causing prolonged droughts in certain areas [5]. Small and marginal farmers make up over 95 % of the farming community, with more than 550 million of the world's 608 million farms being small (less than 2 ha) or medium-sized (less than 50 ha). Therefore, it is essential to boost the adoption rate of precision farming technologies (PFTs) among farmers to ensure sustainable and healthier crop production [6]. Precision farming (PF) technologies can significantly improve land productivity. However, due to the substantial investment required, larger farms with more land and greater access to revenue and capital are more likely to adopt PF. As a result, larger farms are likely to be early adopters of precision farming, leveraging economies of scale to their advantage [7]. For smaller fields, the high cost of positioning systems can be offset by implementing a dead reckoning system (dead reckoning is the process of calculating the current position of a moving object by using a previously determined position, or fix, and incorporating estimates of speed, heading, and elapsed time). This technique, ideal for small, regularly shaped fields, utilizes in-field markers like foam to ensure consistent application, offering farmers a dependable and effective means of spatial field management [8]. Several barriers hinder the adoption of precision technologies among farmers, including low education levels, inexperience, lack of interest, and a general lack of commitment. Many farmers are unaware of the benefits of precision technologies and do not invest time in acquiring the necessary information and knowledge. Additionally, there is often a lack of willingness to cooperate and a very low level of trust among farmers.

Precision agriculture currently encounters several challenges, including unsustainable resource use, prolonged monoculture, intensive animal farming, environmental degradation, uneven digitization, food safety concerns, an inefficient agri-food supply chain, and resistance to change. These problems obstruct the attainment of efficiency, productivity, and sustainability in agricultural production while exacerbating unintended impacts on ecosystems. Additionally, precision agriculture methods often have drawbacks, including high hardware costs, stringent terrain requirements, and susceptibility to environmental factors [9]. Precision farming (PF) greatly reduces greenhouse gas emissions by

utilizing machine guidance and controlled traffic farming to minimize overlaps in farm operations, resulting in a 6 % decrease in fuel consumption. These benefits are even more significant for large-scale fields, offering additional advantages such as decreased soil compaction, runoff, and erosion. PF also decreases effluents, including nitrogen losses in the form of ammonia and nitrogen oxides [10]. To boost productivity, competitiveness, resource use efficiency, and environmental sustainability, farm businesses need to adopt precision farming. Recognized for its ability to achieve these multiple goals, precision farming is an essential tool in modern agricultural practices.

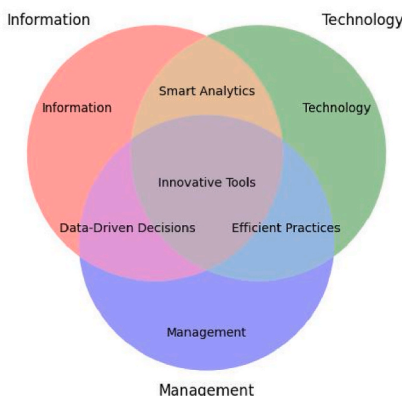
2. AI and IoT integration in smart farming

Smart farming integrates Artificial Intelligence (AI) and the Internet of Things (IoT) into cyber-physical systems for comprehensive farm management. AI applications cover various areas such as soil management, crop health monitoring, disease detection, and weed control. Notable examples include Soil Risk Characterization Decision Support Systems (SRCDSS) (Fig. 2) based on fuzzy logic, Management-Oriented Modeling (MOM), Artificial Neural Networks (ANNs), CALEX, PROLOG, computer vision systems, ANN-GIS, Invasive Weed Optimization (IWO), and Support Vector Machines (SVMs) [11]. A significant AI application involves mobile expert systems, which enable farmers to use smartphones for tasks like disease diagnosis, species identification, and soil health analysis through mobile applications. Additionally, AI facilitates real-time analysis of satellite imagery to monitor farming progress. These innovations provide a scientific basis for precision agriculture, enhancing its effectiveness in optimizing agricultural outputs.

The Internet of Things (IoT) is a network of interconnected devices and technologies that play a pivotal role in precision agriculture and smart farming. IoT architecture in agriculture integrates agricultural sensors with information and communication technology (ICT) and unmanned aerial vehicles (UAVs), facilitating essential data collection for precision farming [12]. The growing presence of IoT and mobile data is crucial in the context of the fourth industrial revolution.

Advances in communication technologies and wireless networks, such as 5G, LoRaWAN, NB-IoT, Sigfox, ZigBee, and Wi-Fi, have significantly broadened the application of IoT across multiple sectors. These technologies enable real-time remote management, high-capacity phenotyping, and improvements in coverage, bandwidth, connection density, and end-to-end latency. In agriculture, IoT technology, integrated with cloud computing, supports smart farming applications like livestock monitoring, smart greenhouses, fishery management, and weather tracking. Common IoT communication technologies include Wi-Fi, Zigbee, LoRa, Bluetooth, and 5G [13]. Communication for agricultural equipment requires attributes like low power consumption,

Main Components of Precision Farming



Agricultural Techniques and Tools

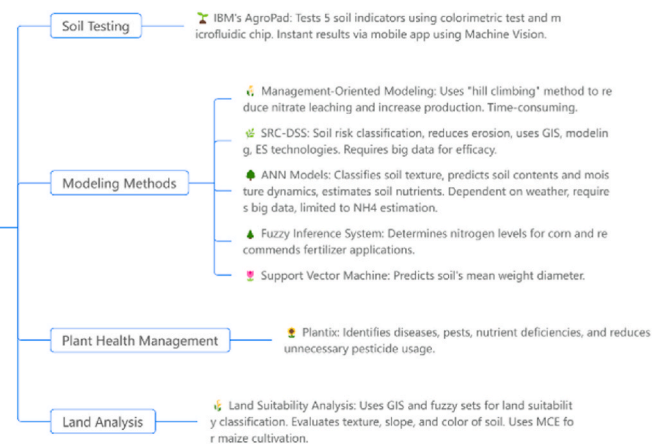


Fig. 1. Fundamentals of precision farming and agricultural tools and techniques useful for precision agriculture.

Components and Relationships in Precision Horticulture

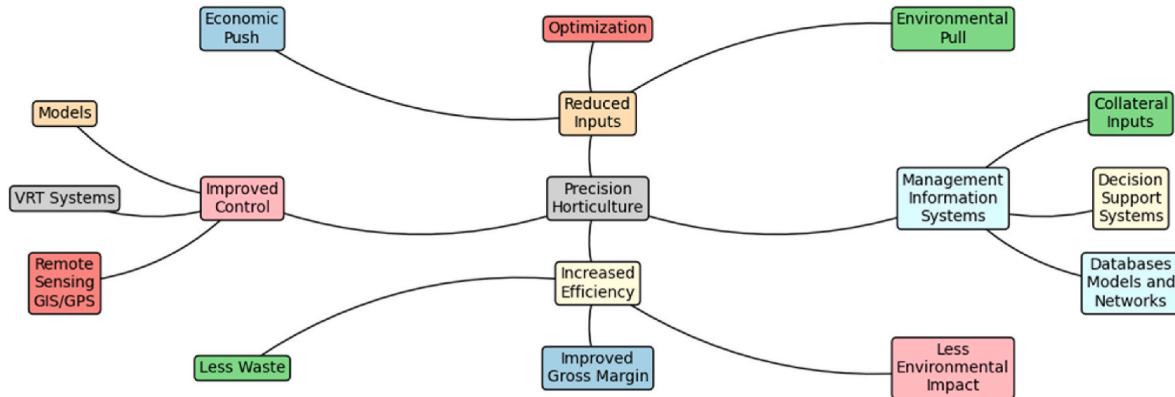


Fig. 2. Elements of precision farming.

cost-effectiveness, and short response times. Among these technologies, LoRa stands out due to its extensive range, low application costs, ease of networking, and capability to support numerous nodes simultaneously.

AI is fundamental to the functionality of Robotics and Autonomous Systems (RAS). Its integration with IoT facilitates the continuous generation of data streams. Data mining techniques are vital for converting agricultural data into actionable insights for decision-making [4]. AI analyzes diverse environmental data and historical farming records stored in large-scale data repositories, uncovering hidden patterns crucial for identifying pests, detecting diseases, predicting yields, and optimizing fertilizer plans within agricultural decision support systems. These insights are essential for improving the precision and efficiency of agricultural practices, enabling well-informed decisions based on comprehensive data analysis (Fig. 3).

3. Technological infrastructure for advanced agriculture

3.1. Wireless Sensor Networks for precision agriculture

Wireless Sensor Networks (WSNs) are composed of stationary or mobile sensor nodes strategically deployed across large areas to collect crucial data essential for precision farming's monitoring, control, and decision-making processes. These nodes integrate physical

environmental data such as temperature, humidity, and other parameters, alongside processing and transmission capabilities, promising broad adoption and versatility in the near future [7]. WSNs are scalable, making installation and expansion straightforward. Research initiatives supported by organizations like the National Science Foundation and Defense Advanced Research Projects Agency (DARPA) focus on developing compact, energy-efficient, and cost-effective sensors tailored for specific agricultural applications while prioritizing environmental sustainability [14].

The integration of WSNs has transformed agricultural practices, benefiting farmers and agricultural experts alike. Modern farm machinery increasingly incorporates sensors, data loggers, and GPS systems to monitor soil conditions, crop health, and field environments. These systems, akin to those developed, encompass vehicles for data collection, management systems, and units for data acquisition and control mounted on farm equipment [4]. They perform detailed field evaluations, collecting information on various factors including soil moisture levels, compaction, fertility, biomass yield, leaf area index, temperature, chlorophyll levels, irrigation status, local climate conditions, pest occurrences, and crop yields [15]. Collected data is transmitted via wired or wireless channels, stored, and analyzed by agricultural professionals to support real-time decision-making systems. Advances in sensor technology have also led to innovations like underground and mobile sensors that enable continuous monitoring of various farming activities.

3.2. Georeferencing technologies in precision agriculture

Georeferencing technologies, such as the Global Positioning System (GPS), utilize satellite constellations to accurately determine locations on Earth, revolutionizing precision agriculture. GPS plays a crucial role in enabling variable rate technology in agricultural practices by providing precise positioning [16]. This capability allows farmers to apply inputs like fertilizers and pesticides according to specific soil conditions. Initially developed by the US Department of Defense, GPS is augmented by other systems like Russia's GLONASS and Europe's Galileo, which is currently in the planning stage and supported by the European Union and the European Space Agency [3].

Geographic Information Systems (GIS) are essential in modern agriculture, integrating computer hardware, software, geographic data, and skilled personnel. GIS effectively captures, stores, updates, manipulates, analyzes, and visualizes geographically referenced information [20]. Beyond basic mapping, GIS encompasses spatial reasoning, data management, and modeling capabilities. It incorporates base maps detailing topography, soil characteristics, nutrient levels, soil moisture, pH

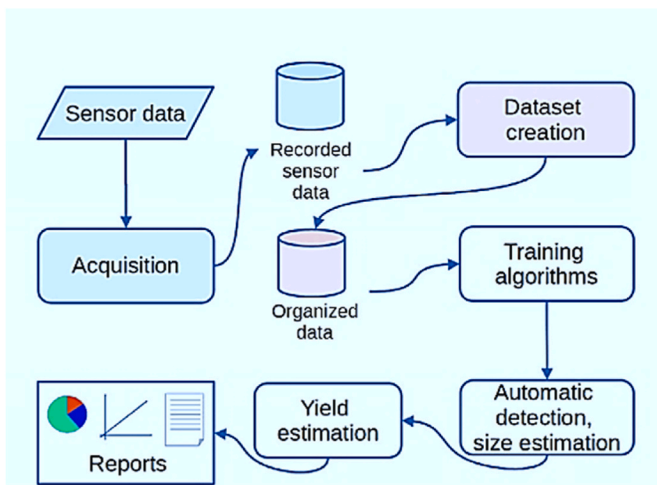


Fig. 3. Data collection and extraction to be applied in precision farming.

levels, fertility, and the intensity of weeds and pests [17]. GIS integrates various data sources and interfaces with decision support tools to precisely apply recommended amounts of nutrients or pesticides.

Guidance systems utilize GPS technology to provide real-time location information for farm equipment and livestock, allowing farmers to enhance operations like planting, harvesting, and herding [11]. Despite obstacles such as limited satellite availability and signal weaknesses, the Global Navigation Satellite System (GNSS) has replaced labor-intensive manual operations with more efficient approaches like Variable Rate Application (VRA) [2]. In earlier agricultural eras, tasks were manual until Agriculture 3.0 introduced mechanized operations with digitalized machinery. As agriculture became more commercialized, efficient management of machinery services became critical to avoid overuse or underuse. GNSS technology is crucial for enhancing the effectiveness and efficiency of agricultural machinery, particularly in the development of fully automated steering systems for tractors [8]. These GNSS-enabled systems not only save time and reduce labor costs but also enhance overall operational efficiency. In precision agriculture, high-resolution navigation solutions are essential for enabling autonomous agricultural robots to perform optimally.

Differential Global Positioning System (DGPS) and remote sensing technologies (Table 1). are revolutionizing precision agriculture by delivering accurate and detailed insights into crop and soil health. DGPS enhances GPS precision by correcting errors observed at known locations, ensuring precise location data crucial for various agricultural applications [5]. Conversely, remote sensing captures electromagnetic radiation and reflection data of crops using aircraft or satellites. This information is critical for monitoring soil conditions, evaluating plant growth, detecting weed outbreaks, and assessing other vital agricultural parameters [18]. Remote sensing operates across wavelengths that include ultraviolet, visible (Red, Green, Blue), and infrared, enabling

farmers to analyze plant vigor and identify issues such as diseases, nutrient deficiencies, and waterlogging effectively.

Remote sensing plays a crucial role in monitoring soil characteristics such as organic matter content and moisture levels through color measurements, supporting holistic soil health management. It also employs vegetation indices like the Normalized Difference Vegetation Index (NDVI) to assess plant health by analyzing the reflectance of near-infrared and red light. Advanced technologies such as Low-Altitude Stationary Surveillance Instrumental Equipment (LASSIE) enable continuous and real-time imaging of crops and soil [19]. These images are integrated with Geographic Information System (GIS) data, enabling precise spatial referencing and enhancing monitoring and management practices in agriculture. Furthermore, integrating sensor data with GNSS coordinates enables the creation of detailed maps such as yield maps and soil maps [6]. These maps are essential for making location-specific decisions in farming, optimizing resource distribution, and implementing efficient water management strategies. Overall, DGPS and remote sensing are pivotal in advancing precision agriculture, fostering sustainable practices, and enhancing overall farm productivity and efficiency.

3.3. The role of broadband connectivity in advancing smart agriculture

Effective broadband connectivity plays a crucial role in overcoming challenges associated with adopting, managing costs, and reducing environmental impact of smart technologies in agriculture. Inadequate connectivity disrupts machine operations, increases human errors, and restricts access to real-time data [20]. This deficiency not only impacts profitability but also impedes the widespread adoption of precision agriculture, which heavily depends on timely information. Farmers and producers equipped with dependable broadband tend to operate more

Table 1
Advanced sensing technologies for precision agriculture.

Sensor Type	Description	Applications	Advantages	Challenges	References
Ultrasound Sensors	Produce high-frequency sound pulses and determine distance by measuring the time interval between emission and echo reception.	Used for distance measurement, obstacle detection, and liquid level detection.	Affordable, small size, suitable for indoor use.	Limited range, affected by environmental factors such as temperature and humidity.	[14]
Time of Flight (ToF) Cameras	Use light sources and detectors to provide 3D distance and intensity measurements.	Used in 3D mapping, gesture recognition, robotics, and industrial automation.	High accuracy, fast frame rate, compact size.	Limited range compared to LiDAR.	[19]
LiDAR (Light Detection and Ranging)	Use laser technology to measure distances and create detailed 3D maps.	Applied in autonomous vehicles, forestry, and precision agriculture for assessing crop structures.	Highly accurate, long-distance capability, effective in dense vegetation.	Expensive, consumes power, requires complex data processing.	[3]
Structured Light Cameras	Project infrared patterns onto scenes and analyze distortion to accurately measure distances.	Used in industrial automation, 3D scanning, and quality inspection.	Provides accurate depth measurement, suitable for indoor environments.	Limited outdoor usability due to interference from ambient light.	[7]
Color Cameras	Capture color and geometric details for detection and characterization tasks.	Used in plant health monitoring, fruit detection, and precision agriculture.	Captures rich data, widely available, cost-effective.	Performance depends on lighting conditions, lacks depth information.	[2]
Thermal Cameras	Measure temperature variations to assess crop health and environmental conditions.	Used in crop diagnosis, irrigation management, and pest detection.	Detects subtle temperature differences, useful day and night.	Lower spatial resolution compared to RGB cameras.	[2]
Multi-spectral Cameras	Measure absorption and reflection across specific electromagnetic spectrum bands to assess physiological conditions.	Used in vegetation health monitoring, nutrient management, and stress detection.	Provides spectral data for specific bands, correlates with plant health indicators.	Costlier than RGB cameras, requires specialized knowledge for interpretation.	[42]
Hyper-spectral Cameras	Measure absorption and reflection across numerous narrow bands to analyze detailed physiological characteristics.	Applied in precision agriculture, remote sensing, and environmental monitoring.	Offers high spectral resolution, identifies subtle vegetation differences.	Expensive, requires intensive data processing, limited sensor availability.	[7]
Laser Sensor-based Spraying	Utilize LiDAR-based laser sensors for precise measurement and spraying applications in orchards.	Used for precision spraying, crop monitoring, and optimizing yield.	Provides high precision, real-time canopy parameter measurements.	Requires significant initial investment, complex integration with existing systems.	[3]
Satellite and Ground-based Reflectance Measurements	Measure plant reflectance to assess crop health and optimize agricultural practices.	Used in remote sensing, precision agriculture, and crop monitoring.	Offers extensive coverage and long-term monitoring capability.	Limited spatial resolution, dependent on weather conditions and satellite availability.	[13]

effectively, underscoring the critical role of robust connectivity in contemporary agriculture. The transformative potential of advanced mobile networks such as 5G and future 6G technologies is increasingly recognized for driving advancements in both business and society [15].

Amid increasing environmental concerns and the pressing need to address climate change, mobile networks play a pivotal role in advancing sustainability and driving innovation across sectors such as smart agriculture, forestry management, biodiversity monitoring, and water resource management. Understanding the capabilities of 5G and anticipating those of 6G networks is essential for meeting the evolving needs of these sectors [4]. For example, in Thailand, telematics-enabled tractors have been developed to assist farmers in efficiently managing their equipment. These advanced systems optimize performance, enhance productivity, and offer additional benefits such as theft prevention, maintenance monitoring, and operational tracking.

4. Advanced technologies in field operations

4.1. Improving precision agriculture through UAV and UGV

Unmanned Aerial Vehicles (UAVs) are aircraft that operate autonomously or under remote control without a human pilot. They have become increasingly valuable in precision agriculture due to their ability to collect data flexibly and with high spatial resolution [21]. UAVs excel in applications such as crop mapping, yield prediction, disease detection, and weed management. Unlike ground-based systems, which face challenges in uneven terrain and accessing remote orchards, UAVs equipped with sensors offer rapid data acquisition capabilities under optimal conditions [13]. This ability allows them to cover large areas swiftly and provide detailed insights into canopy characteristics. In recent years, UAVs have revolutionized agricultural crop phenotyping by offering ultra-high-resolution data acquisition capabilities. Sensors like visible, multispectral, and hyperspectral cameras, along with LiDAR, are mounted on UAVs to measure canopy attributes such as plant height. While LiDAR generates point cloud data of canopies, it may lack detailed feature accuracy [22]. Conversely, high-resolution RGB cameras mounted on UAVs capture images that can be combined to generate intricate models of orchards, providing visual insights into canopy characteristics. This UAV-based approach to canopy measurement is highly beneficial for farmers and researchers, improving precision in spray applications by accurately assessing canopy volume and growth coverage, which is essential for determining seasonal spraying needs [14].

Unmanned Ground Vehicles (UGVs) are equipped with a range of sensory devices, including LiDAR, depth cameras, ultrasonic sensors, and hyperspectral and multispectral cameras. These sensors are essential for collecting the detailed data necessary for UGVs to perform effectively in agricultural tasks. Data fusion techniques are employed to integrate information from these diverse sensors, enhancing the UGV's sensing capabilities and improving its operational accuracy [23]. By combining data from LiDAR, depth cameras, and other sensors, the UGV gains a more precise understanding of its surroundings. In agricultural applications, UGVs equipped with these sensory devices play a crucial role. LiDAR provides comprehensive 3D mapping of terrain, while depth cameras and ultrasonic sensors aid in obstacle detection and navigation [7]. Hyperspectral and multispectral cameras offer valuable insights into crop health and condition, facilitating tasks such as disease detection and monitoring vegetation growth. This integrated approach, utilizing multiple sensors and data fusion techniques, ensures that UGVs are well-equipped to navigate and operate effectively in the varied and challenging environments of agricultural operations [24]. They significantly contribute to enhancing efficiency, precision, and decision-making in modern farming practices.

4.2. Advancements and challenges in Agro-Bot integration for precision agriculture

Agro-Bots are designed for precise agricultural tasks such as harvesting, weeding, spraying, and transportation, leveraging advanced technologies to optimize operations. These innovations ensure accurate chemical application, minimizing environmental impact by reducing harsh chemical use and conserving water. In 2020, the global agricultural robot (Agro-Bot) market reached a value of \$4.9 billion and is expected to grow to \$11.9 billion by 2026, highlighting the increasing adoption and advancement of robotic solutions in agriculture [25]. The primary goal of Agro-Bots is to achieve precise, real-time targeting at the plant level, significantly cutting labor costs, mitigating risks associated with hazardous tasks, and providing farmers with vital data for informed decision-making [19]. However, integrating Agro-Bots presents distinct challenges, particularly in industrialized nations. These challenges include labor shortages and rising production costs. Agricultural fields pose unpredictable conditions such as varying light, weather, and terrain, while crops vary widely in color, size, shape, and sensitivity, often obscured by foliage.

Agro-Bots have successfully been deployed in controlled environments like simulated plantations, greenhouses, and diverse outdoor agricultural settings. Recent advancements in robot technology have enabled automated fruit picking and smart machinery for weed removal among crops [26]. These robots integrate multidisciplinary technologies such as machine vision, electronics, and mechanical engineering. Previously, robotics faced challenges adapting to dynamic environments and performing diverse tasks required in warehouses and fields [17]. Today, sophisticated image processing algorithms coupled with affordable yet powerful sensors and hardware allow flexible task control that accommodates natural objects varying in size and geometry, often appearing as 'unknown' or partially known entities.

4.3. Advancements in autonomous machinery and robotics

Autonomous machinery and robots are rapidly advancing within precision farming (PF), utilizing positioning data and various sensors to perform tasks such as seeding and weed control independently [16]. In greenhouse settings, automation and robotics offer significant benefits, particularly in managing water, fertilizer, heating, CO₂ levels, and ventilation. The history of autonomous agricultural machinery traces back to the early 20th century, with the development of the first tractor steering attachments in 1924 [27]. Initially mechanical, these systems evolved with the integration of computer technology and image sensors in the 1980s, paving the way for machine vision-based guidance systems.

Robotic harvesting systems typically consist of three primary components: a perception module, a manipulator, and an end-effector. The perception module employs onboard sensors like cameras and LiDAR to detect and locate fruits [3]. Once identified, the manipulator is guided to approach and detach the fruit using specialized tools such as grippers or vacuum tubes. Developing such systems requires advancements in multiple fields to achieve seamless integration of these functionalities. These robotic systems aim to optimize harvesting efficiency, reduce labor-intensive tasks, and enhance productivity and precision in agricultural operations.

Automated tractors have a rich history dating back to the early 20th century, marked by the introduction of the first steering attachment for tractors in 1924. Initially relying on mechanical controls until the 1970s, autonomous agricultural machines saw significant advancements in the 1980s with the integration of computers and image sensors, which laid the foundation for machine vision-based guidance systems [28].

Self-driving tractors represent a major leap in agricultural automation, operating autonomously without human intervention. These tractors can determine their position, adjust speeds, and navigate around obstacles such as humans, animals, or objects in the field. They are

equipped with a range of sensors including radar, lasers, and cameras, enabling them to perform tasks such as seeding, planting, and tillage without human intervention [7]. Self-driving tractors excel in path planning and can adapt to operational needs. In some setups, multiple tractors coordinate operations across different fields or collaborate within the same field, often using vehicle-to-vehicle (V2V) communication where one lead tractor with an operator sets the path and speed for others.

Recent advancements in fully autonomous tractor technology are aimed at achieving complete autonomy [1]. This involves integrating advanced sensors like laser or LiDAR for obstacle detection and response, GPS with radio feedback for precise positioning, and sophisticated automation software for optimal path management and operational control [29]. These technological strides not only enhance operational efficiency but also reduce labor demands and optimize farming practices through comprehensive automation.

In a national project focusing on precision farming, a semi-autonomous robot harvester is being developed for specific tasks:

1. **Mechatronic Agriculture Vehicle:** This vehicle navigates autonomously through vineyard rows to perform scouting operations, under remote monitoring and control by a farmer.
2. **Robotic Arm:** Integrated onto the agricultural vehicle, a multi-axis robotic arm facilitates image acquisition. Positioned near the arm's end effector, a camera captures images of diseased leaves during inspections among vineyard rows [21]. These images are stored in the vehicle's computer memory and compared against reference images.
3. **Control System:** Through image processing, if a captured image matches any in the database, it identifies the corresponding disease. Upon diagnosis, a control signal triggers a relay to open a solenoid valve connected to the pesticide storage tank.
4. **Pesticide Spraying Unit:** Once a disease is confirmed, the solenoid valve opens to allow pesticide flow through a pump. Pressurized pesticides are then sprayed through nozzles for a specified duration, targeting identified diseased leaves with appropriate pesticides in accordance with NRCG guidelines [4].

4.4. Monorail transporters in mountain agriculture

Mountainous and hilly regions are vital for agriculture, producing a range of crops despite their challenging terrain and narrow, uneven roads that hinder traditional machinery designed for flatlands. This disparity often results in machinery availability issues or reduced performance. To tackle these obstacles, monorail transporters have emerged as a key solution for mechanized operations in mountainous orchards [30]. These transporters are valued for their smooth operation, strong climbing ability, and capacity to maneuver narrow paths with a tight turning radius. Despite their benefits, expanding the monorail track network presents new challenges such as diverse terrains, dense canopies, and varying shading conditions, complicating real-time monitoring and control.

To effectively tackle these challenges, integrating autonomous operation and navigation systems into monorail transporters is crucial. This aligns with the shift towards intelligent agriculture, enabling these machines to adjust to dynamic environments and improve operational efficiency. Looking ahead, monorail transporters can facilitate cooperative operations with different agricultural machinery [9]. For example, integrated sprayers can autonomously apply treatments at optimal times, thereby enhancing spray effectiveness and operational efficiency. Such advancements not only enhance the relevance of monorail transporters in agriculture but also drive the evolution of intelligent farming practices.

5. Advancements in precision farming for orchard and vineyard management

5.1. Advancements in 3D fruit localization for robotic harvesting in orchards

Recent developments in deep learning have markedly improved the precision of fruit detection, allowing for reliable identification even in challenging and intricate surroundings. However, the development of 3D fruit localization, crucial for robotic harvesting, has faced challenges [15]. Challenges such as complex fruit shapes, varying orientations, clustering, changing light conditions, and foliage obstructions have hindered the effectiveness of current sensors in accurately locating fruits in natural orchard environments [31]. The Active Laser-Camera Scanning (ALACS) system addresses these obstacles by integrating depth cameras and laser scanning technologies [7]. ALACS has demonstrated superior performance in precisely identifying apples within orchards. During tests involving a harvesting robot equipped with a vacuum-based end effector, ALACS successfully harvested fruits with high efficiency.

5.2. Tree canopy reconstruction and fruit localization

In modern agriculture, understanding how fruits are distributed spatially within tree canopies is crucial. This information extends beyond simply counting fruits; it enables accurate predictions of harvest timing, which in turn supports detailed planning and effective allocation of resources [24]. As automation becomes more prevalent in agriculture, robots specialized for tasks like fruit picking depend significantly on spatial data [32]. Accurate information about fruit locations allows these robots to streamline their operations, reducing fruit damage and navigating orchards efficiently.

5.3. Optimizing mango yield forecasting and detection through advanced deep learning techniques

MangoYOLO is an innovative architecture tailored for mango fruit detection in orchards, boasting real-time processing capabilities with just 33 layers, a significant reduction from YOLOv3's 107 layers [29]. Despite its streamlined design, MangoYOLO achieves comparable or superior accuracy compared to other detectors designed for this specific task. Recent advancements in image processing and deep learning models have proven pivotal for improving yield estimation in agriculture [33]. Machine learning techniques such as Artificial Neural Networks and Decision Trees have also demonstrated effectiveness in forecasting yields well before harvest, offering valuable insights for decision-making and enhancing overall productivity. By leveraging data on soil conditions, climate patterns, weather forecasts, soil health, and historical crop yields, predictive models can accurately project crop yields for the upcoming growing season. This holistic approach enables farmers to optimize farming practices and enhance overall yield management effectively.

5.4. The PACMAN SCRI project-enhanced digital technologies for precision management of apple crop loads

The PACMAN SCRI project, focusing on Digital Technologies for Precision Apple Crop Load Management (PACMAN), has shown over many years that this approach is highly effective in managing crop load. Effective crop load management directly impacts yield, fruit quality, size, return bloom, and ultimately, the profitability of orchards [34]. These activities include three main practices: pruning, chemical thinning, and hand thinning, which have been extensively covered in earlier publications. Through a 4-year national project supported by USDA--NIFA SCRI, we are further developing and improving regional PACMAN guidelines.

New tools leveraging computer vision, artificial intelligence (AI),

and machine learning (ML) are increasingly advancing PACMAN techniques. These tools are designed to enhance the efficiency and speed of measuring fruitlets, predicting fruit set using growth rate models, and providing precise forecasts for harvest [27]. While the tested tools like the Malusim app, Ferri app, and Farm Vision (Pometa) scanning demonstrated differing levels of accuracy, they represent notable progress in the precise management of apple crop loads.

5.5. MATLAB-based mechatronics system for automated grape leaf health monitoring and disease management

A mechatronics system based on MATLAB (Fig. 4) is proposed for monitoring grape leaf health using histogram analysis. This system detects early signs of major grape diseases like Downy Mildew, Powdery Mildew, and Anthracnose, alerting farmers promptly [35]. Operating under the principles of Precision Farming, it automatically initiates pesticide spraying when necessary. This method optimizes pesticide usage economically and frees up farmers' time for other critical agricultural activities [19]. Despite Maharashtra's substantial annual grape production, the adoption of automated systems for grape orchard management remains limited. This gap presents an opportunity to enhance efficiency and productivity in grape farming practices through the integration of advanced automated technologies.

5.6. Project PANTHEON: IoT-Integrated SCADA system for precision management of hazelnut orchards

Project PANTHEON is dedicated to developing a Supervisory Control and Data Acquisition (SCADA) system designed specifically for precision farming in hazelnut orchards. The project aims to establish a novel real-time software framework inspired by industrial SCADA systems (Fig. 5), customized to meet the needs of agricultural applications [9]. PANTHEON is developing an integrated system featuring an IoT-based agrometeorological monitoring network. This setup includes a weather station to collect meteorological data and several soil moisture probe nodes for monitoring soil humidity and temperature [17]. Moreover, both terrestrial and aerial robots traverse the orchard, employing sensors like LiDAR and multispectral cameras to gather precise measurements at the level of individual trees. The SCADA system being developed as part of the PANTHEON project integrates all these

components, offering an advanced and efficient solution for precisely managing hazelnut orchards.

5.7. BYELAB: A rototype mobile robot for close-range sensing and precision agriculture

The ByeLab, or Bionic eYe Laboratory, is a mobile robotic lab created at the Faculty of Science and Technology of the Free University of Bolzano. This cutting-edge system functions as a semi-autonomous robot engineered for monitoring the health of orchards and vineyards. Its primary function involves using LIDAR technology and specialized algorithms to assess and measure the shape and volume of orchard canopies [10]. A mobile vehicle used for plant sensing must meet specific technical criteria: (a) it needs to maneuver easily within orchards, navigate off-road terrain, and execute turns on steep slopes; (b) it should be conveniently transportable to various investigation sites; and (c) it must support the integration of crop monitoring sensors through a custom-designed structure [32]. Additionally, the robotic system must maintain a high level of safety and reliability to ensure the protection of nearby humans and the environment.

5.8. Advanced sensor integration & data tools for precision ag

The PAGFRUIT research project has created two main tools: AK_ACQS and AK_FRAEX. AK_ACQS, the acquisition stage software, allows for the simultaneous operation of multiple sensors like time-of-flight (ToF) and stereo vision cameras, alongside a GNSS receiver [18]. It is designed for data collection in fruit orchards using sensors mounted on terrestrial vehicles. AK_ACQS coordinates computers and sensors by sending commands remotely through a graphical user interface (GUI), and it enhances integration by adding an abstraction layer to each sensor's library stack [22].

On the other hand, AK_FRAEX, the extraction software, processes videos captured with the ToF camera to generate datasets. The chosen ToF camera is the Azure Kinect DK, renowned for its RGB-D capabilities suitable for commercial and outdoor agricultural applications [24]. The camera provides multiple types of information including color images, IR images, depth data, and IMU motion data. In this project, the stereo vision camera ZED 2 and the GNSS receiver SimpleRTK2B Basic Starter Kit complement the setup. AK_FRAEX serves as a desktop tool for processing data after field acquisition, enabling the extraction of frames and generation of metadata from MKV-format videos captured with the Azure Kinect camera [11]. Users can configure initial parameters through a GUI to automate these processes and manage image sets effectively.

6. Emerging technologies in precision agriculture

6.1. High-throughput phenotyping for precision agriculture

High-throughput phenotyping (Fig. 6) is transforming precision agriculture by facilitating fast and precise assessment of both quantitative and qualitative plant traits. This approach is pivotal in expediting crop breeding initiatives [36]. Leveraging state-of-the-art technologies such as remote sensing, spectral imaging, and robotics, comprehensive data on diverse plant attributes such as growth rate, yield capacity, disease resilience, and morphology are collected. Farmers leverage this data to gain valuable insights into crop performance, guiding decisions on fertilization, irrigation, harvest timing, and pest management strategies [37]. Moreover, high-throughput phenotyping supports breeders in developing new crop varieties adapted to local environmental conditions, aiming for higher yields. Extensive datasets gathered from visible and near-infrared hyperspectral imaging empower machine learning methods such as least squares regression (LSR) to forecast precise biochemical and physicochemical traits, surpassing conventional vegetative indices [28].

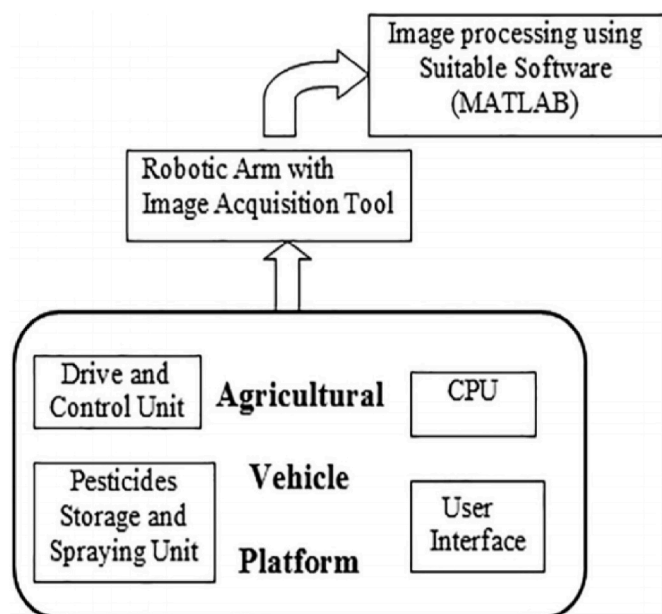


Fig. 4. Working module of MATLAB Mechatronics System.

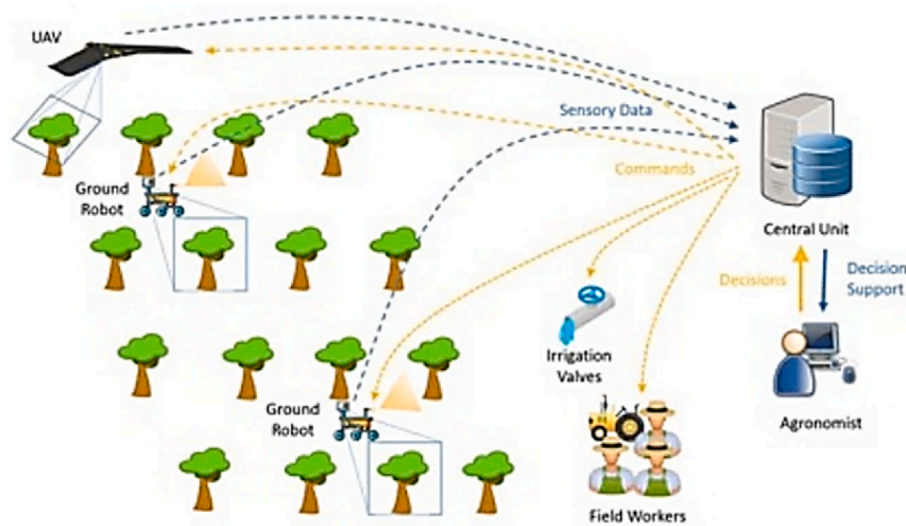


Fig. 5. Working module of SCADA system.

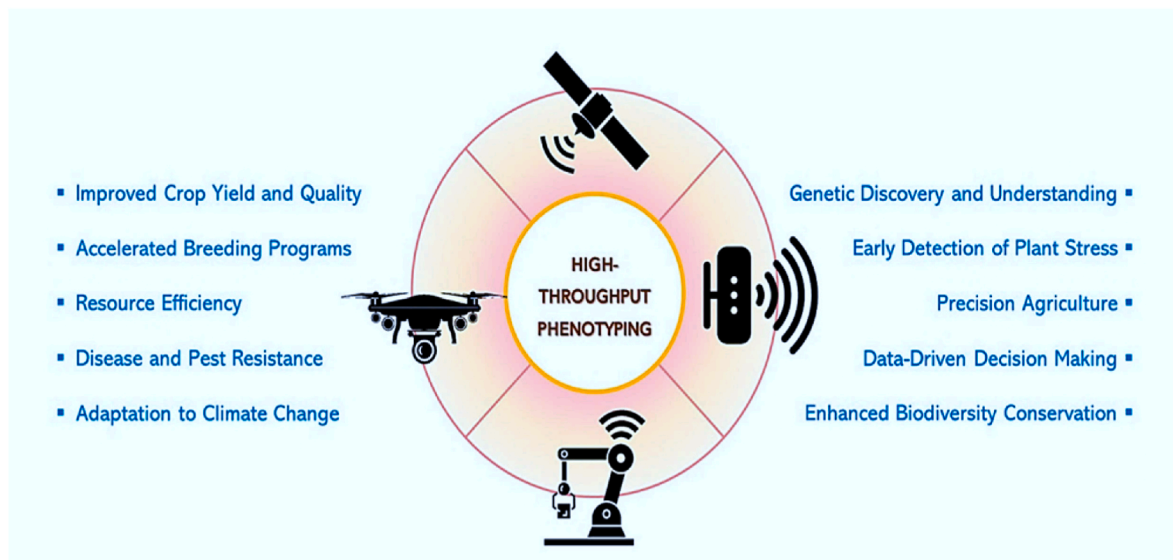


Fig. 6. The significance of high-throughput phenotyping in agriculture.

Agricultural phenotyping involves various critical elements: a) Structural Analysis: Precise determination of metrics like canopy volume, plant height, leaf area coverage, and biomass greatly enhances agricultural operations. b) Plant/Fruit Detection: Automation tasks like pruning, harvesting, and seeding rely on precise object detection within the environment. c) Physiology Assessment: The interaction between the canopy and sunlight generates distinct spectral signatures that indicate the plant's physiological status. Different indices derived from crop spectral responses evaluate parameters such as nitrogen deficiency, chlorophyll concentration, water stress, and pest infestation [21]. Advanced sensing devices like Infrared gas analyzers directly measure multiple physiological plant parameters.

6.2. Variable Rate Technologies (VRTs)

Variable rate technologies (VRTs) enable precise application of inputs like nitrogen fertilizers, tree spraying, or center pivot irrigation tailored to specific locations within fields. Conventional farm equipment can integrate electronic control units (ECUs) and onboard GPS systems

to effectively implement VRTs [38]. For example, spray booms and spinning disc applicators equipped with ECUs and GPS are utilized for targeted spraying (Table 2) in designated areas, improving the management of field zones. Variable Rate Technologies (VRT) aim to enhance input management, leading to either decreased costs and environmental effects or improved yields and product quality. Two main approaches in VRT applications are map-based and sensor-based. Map-based VRT utilizes historical data and prescription maps generated by GIS to adjust fertilizer application, seeding rates, and pesticide application rates [27]. This method utilizes process control technologies to ensure precise input management. In contrast, sensor-based VRT adjusts application rates in real-time based on immediate crop or soil characteristics detected by onboard sensors. This approach allows for dynamic adjustments during operations, appealing to farmers for its adaptability and responsiveness to field conditions. Both map-based and sensor-based systems have their own advantages and drawbacks. Farmers often prefer sensor-based systems for their real-time responsiveness [7]. However, a combination of both approaches shows potential for maximizing benefits in future agricultural practices.

Table 2
Innovations and strategies in orchard spraying technology.

Topic	Description	References
CASA Sprayer	Advanced orchard sprayer (ISAFRUIT CASA) designed for precise, efficient, and environmentally safe spray applications tailored to crop needs.	[46]
Nozzles	AI nozzles equipped with canopy sensors may encounter airflow delays; minimal residue level differences observed between fine hollow cone and coarse air induction nozzles.	[47]
Dose Rates	Modern orchards support reduced dose rates with sufficient leaf coverage; Italian researchers achieved a 20 % reduction in doses without loss of efficacy.	[1]
Air Volume and Speed	Optimal spray deposition requires 25–30,000 m ³ /hr ⁻¹ air volume; modern sprayers often operate at 40–60,000 m ³ /hr ⁻¹ . Posing a challenge for axial fan sprayers.	[6]
Canopy Sensors	Canopy sensors, whether infrared or ultrasound-based, identify the presence, height, and density of the canopy to optimize spray application volumes. Among these, infrared sensors are noted for their reliability and cost-effectiveness.	[4]
Patternator Designs	Patternators capture spray to visualize and adjust spray patterns according to tree shapes; reduces spray volumes for improved canopy penetration and efficiency.	[34]
Soil Mapping	Veris Technologies' tool utilizes GPS to map soil electrical conductivity (EC) for precision agriculture, identifying field soil variations.	[34]
Yield Mapping	Essential for assessing orchard yield variability; challenges remain in accurately measuring variability within orchards despite widespread industry practice.	[20]

6.3. Blockchain technology in precision agriculture

Blockchain operates as a decentralized and distributed database that maintains a growing sequence of ordered records or blocks, originally introduced in the context of cryptocurrency. It provides transparency, immutability, and data reliability, fostering trust among participants in supply chains. In precision agriculture, blockchain facilitates the integration of digital technologies by overcoming corporate barriers. This development tackles technical hurdles in smart farming by facilitating remote monitoring and control of agricultural equipment, exemplified by systems such as the “IoT applied Greenhouse Monitoring System” [39]. One significant issue blockchain addresses is the inadequacy and insecurity of infrastructure for sharing agricultural data. It also tackles delays in remote-sensing satellite data for monitoring cropland variability. Blockchain offers solutions to these challenges through its decentralized, anonymous, and secure framework [17]. The goal is to offer lightweight, distributed, and transparent security and privacy solutions for IoT applications in smart farming.

As blockchain technology advances, it proves to be a dependable, efficient, and secure platform for monitoring farm operations. It improves the organization of diverse agricultural tasks such as irrigation scheduling, energy management, machinery integration, labor allocation, and coordination of robot teams and autonomous UAVs [34]. This capability is crucial in the food supply chain, addressing concerns related to food safety and mitigating issues arising from asymmetric and fragmented information.

6.4. Image segmentation and image processing algorithms for agricultural automation

Image segmentation plays a crucial role in agricultural automation by distinguishing plants from backgrounds and accurately identifying fruits amidst foliage and other environmental elements. This capability is essential for automating tasks like harvesting, estimating yields, and

detecting diseases [40]. While advanced image processing technologies enable rapid and precise detection, challenges such as varying lighting conditions can affect the performance of color-based segmentation methods. Particularly challenging are situations where fruits, like green apples or grapes, blend with their surroundings, which often include leaves and branches, sharing similar colors [9]. To overcome these challenges, advanced segmentation techniques have emerged. These methods go beyond traditional color-based approaches by incorporating features such as texture and shape and considering pixel relationships within groups. By doing so, they enhance segmentation accuracy even in difficult conditions [26]. These advancements contribute to more dependable and efficient automation across agricultural applications.

Image processing presents a dependable, effective, and exact substitute for manual methods in evaluating crop quality. It utilizes sophisticated algorithms for recognizing patterns, ensuring precise assessments swiftly and straightforwardly [13]. In the realm of harvesting robots functioning in natural settings with fluctuating light conditions, the precision of image segmentation algorithms assumes critical importance. Sunlight frequently causes shadows in images, potentially misleading the system's diagnostic accuracy.

6.5. Developing robust segmentation systems

To tackle the challenges posed by varying sizes of plants and fruits, as well as fluctuating sunlight levels, it is crucial to develop a robust segmentation system capable of real-time operation under all environmental conditions [15]. A novel color space, designed specifically for agricultural environments, successfully distinguishes green plants from their backgrounds under varying light conditions and growth stages. The machine vision system proposed operates in two stages: firstly, it detects color conditions [41]. and subsequently uses a hybrid artificial neural network-harmony search (ANN-HS) classifier for segmentation. This method ensures precise and dependable segmentation across a wide range of environmental scenarios, thereby enhancing the efficiency of automated agricultural operations.

6.6. Endo-therapy: advancing sustainable management in olive orchards

Endo-therapy, a method for delivering active substances directly into the vascular system of plants through trunk injection, represents a significant advancement in sustainable management practices for olive crops. This technology allows for the treatment of vascular diseases that cannot be effectively addressed through traditional foliar applications, ensuring precise delivery without the risk of unintended drift [42]. Advances in controlled release formulations further enhance the precision and sustainability of crop protection efforts.

Trunk injection involves the direct introduction of active ingredients into a plant's vascular system, surpassing conventional methods like soil drenches or trunk sprays, especially effective for trees. Initially developed for high-value ornamental plants, commercial trunk injections have evolved to cater to both monocot and dicot plants using various techniques, including drill-based and drill-free methods with or without pressure [43]. Today, endo-therapy is widely utilized to combat diseases and pests residing within tree trunks, ensuring targeted treatment delivery to affected tissues. Moreover, it serves as a viable alternative to foliar sprays in environments where minimizing exposure risks is critical, such as urban settings.

Historically, tree injections were pioneered in the early 20th century to manage diseases like chestnut blight canker using metal and salt solutions. Despite challenges in mechanizing olive farming, precision endo-therapy offers innovative solutions for effectively managing both biotic and abiotic stresses. Recent studies demonstrate its efficacy in delivering active substances into the systemic tissues of mature olive trees using minimal injection volumes [14]. This capability opens new avenues for treating vascular and foliar diseases, applying micro-nutrients, and administering bio-stimulants to promote overall tree

health while reducing environmental impacts. Integrated into smart agricultural systems, endo-therapy stands out as a transformative technology for olive orchards, paving the way for integrated pest management in tree cultivation.

6.7. Reducing pesticide use and its environmental impact

Global agriculture relies heavily on pesticides, with over three million tonnes used annually, which poses significant risks to biodiversity, the environment, and human health. Thus, reducing pesticide use and adopting sustainable practices are critical. Precision farming (PF) technologies offer effective solutions to minimize pesticide usage in crop management [28]. In recent years, these technologies have been increasingly applied in pest management, providing methods to target specific areas affected by pests rather than treating entire fields. This approach, facilitated by tools like multispectral and hyperspectral cameras, canopy sensors, weather stations, and smart traps, enables precise and timely interventions to control pests, thereby reducing overall pesticide application [7]. The agricultural sector faces substantial challenges in balancing high food production with minimizing environmental impacts such as pollution, biodiversity loss, and greenhouse gas emissions. Precision farming technologies present promising strategies to address these challenges by promoting sustainable agricultural practices. Ensuring precise pesticide application through technologies like Automatic Navigation (AN) is crucial for environmental protection, especially in orchards [20]. Integrating AN with Precision Variable-Rate Spraying (PVS) can further enhance efficiency and reduce pesticide usage by tailoring spraying decisions based on fruit tree canopy characteristics. Currently, AN and PVS integration is not fully optimized, highlighting the potential for improvement in pesticide application practices.

7. Advances in automated Harvest support and plant Health monitoring in agriculture

7.1. Harvest support approaches

Harvesting is a crucial task in agriculture, involving the gathering of ripe crops or fruits from fields. It traditionally relies on labor-intensive methods to detect and collect mature crops/fruits [44]. Manual sorting based on criteria such as size, maturity, shape, and damage typically follows harvesting, but this process is slow and prone to human error, leading to variability in product quality. Hence, there is a clear need for more accurate and efficient methods to assess harvested fruits or crops [37]. Automated sorting systems offer significant improvements over manual methods in terms of speed, reliability, and precision. Fruit counting is also essential to facilitate harvesting operations efficiently.

7.2. Fruit grading and ripeness detection

Fruit grading involves sorting fruits based on characteristics such as size, shape, and maturity. Manual grading is labor-intensive and prone to human error. Machine vision technology addresses these challenges by automating the process and ensuring precise fruit grading [28]. As an illustration, a mango grading system utilizes an image processing algorithm that integrates region-based global thresholding color binarization, median filtering, and morphological analysis. This approach effectively categorizes mangos into three mass grades (large, medium, and small) in real-time with high accuracy [33]. Furthermore, this technology is adaptable and can be adjusted for other crops with appropriate modifications.

7.3. Fruit counting and yield prediction

Accurate fruit counting is crucial for estimating yield, a task that poses challenges for machine vision technology. Counting and sizing

fruits via machine vision requires ensuring all fruits on a tree are visible and not obscured by leaves. However, identifying each fruit in tree images can be scientifically challenging, especially early in the season when some fruits are hidden within the canopy [45]. Variations in fruit appearance due to lighting conditions and occlusion by foliage or other fruits may lead to inaccuracies in counting. Nevertheless, early yield prediction is vital for effective planning of labor, storage, harvest management, transportation, grading, and storage logistics.

7.4. Approaches for protecting plant and fruit health and detecting diseases

The health of plants and fruits faces multiple threats, with weeds posing a significant challenge as they compete with plants for vital resources such as nutrients, light, and water, thereby impacting growth and product quality if not promptly identified and removed [26]. Machine vision technology has been extensively applied to automate weed detection, especially in scenarios where there is minimal color contrast between the plant and the weed [46]. Additionally, insects represent another critical threat to plant health. Automated insect monitoring helps farmers protect their crops and fruits. Machine vision systems have also proven valuable in detecting diseases and identifying nutrient deficiencies.

Detection of Insects: Managing pests is a top priority for farmers due to their potential to reduce crop yields and cause damage. Consequently, farmers employ various strategies to mitigate pest impact. In recent years, there has been a reliance on chemical pesticides due to their initial cost-effectiveness, accessibility, rapid action, and limited farmer knowledge, despite the associated risks to public health, animals, and the environment [37]. Therefore, exploring alternative pest control methods to chemical pesticides is crucial, considering the significant damage insects can inflict on plants and fruits.

Detection of Diseases and Nutrient Deficiencies: Automatically identifying unhealthy plant parts is a critical and complex research area. Detecting diseases (Fig. 7) and nutrient deficiencies is challenging due to the variability in symptoms depending on the specific disease, deficiency type, fruit, or plant species [47]. Diseased plant parts exhibit changes in color and texture, which can be further influenced by varying light conditions.

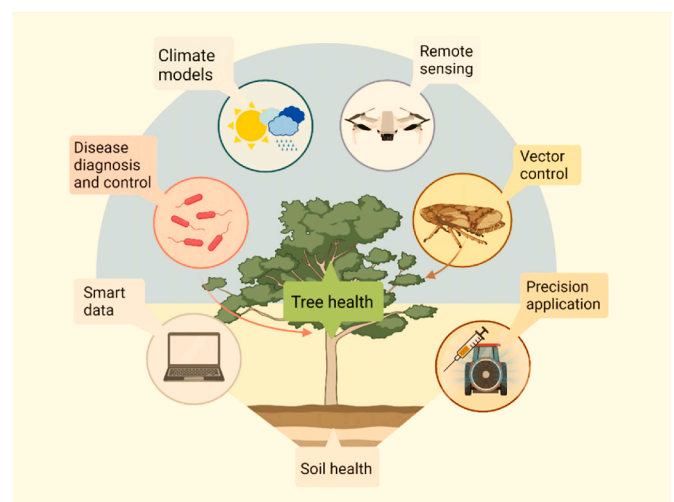


Fig. 7. Future orchard cultivation, focusing on tree health, can be effectively managed by integrating various innovations. These include automated soil mapping, disease diagnostics, alternative pest control methods, climate forecasting, remote sensing of biophysical parameters, strategic vector control, and precise application techniques like controlled release formulations and endo-therapy. Growers can oversee these technologies in real-time using advanced data management systems.

8. Conclusion

The integration of Artificial Intelligence (AI) and the Internet of Things (IoT) in precision agriculture offers transformative benefits for crop monitoring and management. Through advanced technologies such as image processing, machine vision, and spectral imaging, enhanced capabilities in fruit detection, yield prediction, and disease monitoring have been achieved. Wireless Sensor Networks (WSNs) provide crucial real-time environmental data, enabling informed decision-making and resource optimization. Autonomous robots streamline agricultural tasks like harvesting, weeding, and spraying, reducing labor costs and environmental impact through precise chemical application. The synergy of AI and IoT optimizes productivity, reduces environmental footprint, and supports sustainable farming practices essential for meeting global food demands. These innovations empower farmers with smarter, data-driven solutions that enhance crop management efficiency and promote responsible agricultural development. As technology continues to evolve, the future holds promise for further advancements in precision agriculture, shaping a more efficient, productive, and sustainable agricultural landscape.

CRediT authorship contribution statement

Kushagra Sharma: Data curation, Validation, Visualization, Writing – original draft. **Shiv Kumar Shivandu:** Conceptualization, Writing – review & editing.

References

- [1] J.R. Rosell-Polo, F.A. Cheein, E. Gregorio, D. Andújar, L. Puigdomènech, J. Masip, A. Escolà, Advances in structured light sensors applications in precision agriculture and livestock farming, *Adv. Agron.* 133 (2015) 71–112, <https://doi.org/10.1016/b.s.agron.2015.05.002>.
- [2] S. Haneklaus, H. Lilienthal, E. Schnug, 25 years precision agriculture in Germany—a retrospective, in: 13th International Conference on Precision Agriculture, 2016, July, <https://doi.org/10.1201/NOE0849338304.ch310>.
- [3] N. Khardia, S. Sharma, H. Kumawat, Precision farming, *A Monthly Peer Reviewed Magazine for Agriculture and Allied Sciences* 63 (2022), <https://doi.org/10.60151/envec/zblx2641>.
- [4] S.M. Say, M. Keskin, M. Sehri, Y.E. Sekerli, Adoption of precision agriculture technologies in developed and developing countries, *The Online J. Sci. Technol.* January 8 (1) (2018) 7–15, <https://doi.org/10.1080/00103624.2021.1900862>.
- [5] T. Anand, S. Sinha, M. Mandal, V. Chamola, F.R. Yu, AgriSegNet: deep aerial semantic segmentation framework for IoT-assisted precision agriculture, *IEEE Sensor. J.* 21 (16) (2021) 17581–17590, <https://doi.org/10.1109/jsen.2021.3071290>.
- [6] E. Mavridou, E. Vrochidou, G.A. Papakostas, T. Pachidis, V.G. Kaburlasos, Machine vision systems in precision agriculture for crop farming, *J. Imaging* 5 (12) (2019) 89, <https://doi.org/10.3390/jimaging5120089>.
- [7] S. Jain, S. Bhujel, U. Shrivastava, A. Mohapatra, G. Mishra, Advancements in drone technology for fruit crop management: a comprehensive review, *Int. J. Envir. Clim. Change* 13 (11) (2023) 4367–4378, <https://doi.org/10.9734/ijec/2023/v13i113617>.
- [8] F.Y. Narvaez, G. Reina, M. Torres-Torriti, G. Kantor, F.A. Cheein, A survey of ranging and imaging techniques for precision agriculture phenotyping, *IEEE ASME Trans. Mechatron.* 22 (6) (2017) 2428–2439, <https://doi.org/10.1109/tmech.2017.2760866>.
- [9] L. Giustarini, S. Lamprecht, R. Retzlaff, T. Udelhoven, N.B. Rossello, E. Garone, A. Gasparri, PANTHEON: SCADA for precision agriculture, in: *Handbook of Real-Time Computing*, Springer Nature Singapore, Singapore, 2022, pp. 641–679, https://doi.org/10.1007/978-981-287-251-7_42.
- [10] R. Vidoni, R. Gallo, G. Ristorto, G. Carabin, F. Mazzetto, L. Scalera, A. Gasparetto, Byelab: an agricultural mobile robot prototype for proximal sensing and precision farming, in: *ASME International Mechanical Engineering Congress and Exposition*, vol. 58370, American Society of Mechanical Engineers, 2017, November V04AT05A057, <https://doi.org/10.1115/imece2017-71216>.
- [11] E.M.B.M. Karunathilake, A.T. Le, S. Heo, Y.S. Chung, S. Mansoor, The path to smart farming: innovations and opportunities in precision agriculture, *Agriculture* 13 (8) (2023) 1593, <https://doi.org/10.3390/agriculture13081593>.
- [12] L. Tomaszewski, R. Kolakowski, M. Zagóda, Application of mobile networks (5G and beyond) in precision agriculture, in: *IFIP International Conference on Artificial Intelligence Applications and Innovations*, Springer International Publishing, Cham, 2022, June, pp. 71–86, https://doi.org/10.1007/978-3-031-08341-9_7.
- [13] M.T. Pratama, S. Kim, S. Ozawa, T. Ohkawa, Y. Chona, H. Tsuji, N. Murakami, Deep learning-based object detection for crop monitoring in soybean fields, in: *2020 International Joint Conference on Neural Networks (IJCNN)*, IEEE, 2020, July, pp. 1–7, <https://doi.org/10.1109/ijcnn48605.2020.9207400>.
- [14] X. Feng, F. Yan, X. Liu, Study of wireless communication technologies on Internet of Things for precision agriculture, *Wireless Pers. Commun.* 108 (3) (2019) 1785–1802, <https://doi.org/10.1007/s11277-019-06496-7>.
- [15] D. Thakur, Y. Kumar, A. Kumar, P.K. Singh, Applicability of wireless sensor networks in precision agriculture: a review, *Wireless Pers. Commun.* 107 (2019) 471–512, <https://doi.org/10.1007/s11277-019-06285-2>.
- [16] P.C. Pandey, A.K. Tripathi, J.K. Sharma, An evaluation of GPS opportunity in market for precision agriculture, in: *GPS and GNSS Technology in Geosciences*, Elsevier, 2021, pp. 337–349, <https://doi.org/10.1016/b978-0-12-818617-6.00016-0>.
- [17] P.K. Mani, A. Mandal, S. Biswas, B. Sarkar, T. Mitran, R.S. Meena, Remote sensing and geographic information system: a tool for precision farming, *Geospatial Technologies for Crops and Soils* (2021) 49–111, https://doi.org/10.1007/978-981-15-6864-0_2.
- [18] J.C. Padro, F.J. Muñoz, J. Planas, X. Pons, Comparison of four UAV georeferencing methods for environmental monitoring purposes focusing on the combined use with airborne and satellite remote sensing platforms, *Int. J. Appl. Earth Obs. Geoinf.* 75 (2019) 130–140, <https://doi.org/10.1016/j.jag.2018.10.018>.
- [19] M. Vazquez-Arellano, H.W. Griepentrog, D. Reiser, D.S. Paraforos, 3-D imaging systems for agricultural applications—a review, *Sensors* 16 (5) (2016) 618, <https://doi.org/10.3390/s16050618>.
- [20] U. Shafi, R. Mumtaz, J. García-Nieto, S.A. Hassan, S.A.R. Zaidi, N. Iqbal, Precision agriculture techniques and practices: from considerations to applications, *Sensors* 19 (17) (2019) 3796, <https://doi.org/10.3390/s19173796>.
- [21] P. Velusamy, S. Rajendran, R.K. Mahendran, S. Naseer, M. Shafiq, J.G. Choi, Unmanned aerial vehicles (UAV) in precision agriculture: applications and challenges, *Energies* 15 (1) (2021) 217, <https://doi.org/10.3390/en15010217>.
- [22] J. Yuan, W. Ji, Q. Feng, Robots and autonomous machines for sustainable agriculture production, *Agriculture* 13 (7) (2023) 1340, <https://doi.org/10.3390/agriculture13071340>.
- [23] Q. Vu, M. Raković, V. Delic, A. Ronzhin, Trends in development of UAV-UGV cooperation approaches in precision agriculture, in: *Interactive Collaborative Robotics: Third International Conference, ICR 2018, Leipzig, Germany, September 18–22, 2018, Proceedings 3*, Springer International Publishing, 2018, pp. 213–221, https://doi.org/10.1007/978-3-319-99582-3_22.
- [24] A. Sassu, V. Psiroukis, F. Bettucci, L. Ghiani, S. Fountas, F. Gambella, Unmanned aerial system plant protection products spraying performance evaluation on a vineyard, *Precis. Agric.* (2024) 1–31, <https://doi.org/10.1007/s11119-024-10155-8>.
- [25] S. Singh, P.S. Chauhan, C. Sethi, A. Jain, S.K. Agnihotri, Design and development of multipurpose agriculture robot (Agrobot), in: *Distributed Intelligent Circuits and Systems*, 2024, pp. 389–420, https://doi.org/10.1142/9789811279539_0011.
- [26] D. Sarri, S. Lombardo, R. Lisci, V. De Pascale, M. Vieri, Agrobot smash a robotic platform for the sustainable precision agriculture, in: *Innovative Biosystems Engineering for Sustainable Agriculture, Forestry and Food Production: International Mid-term Conference 2019 of the Italian Association of Agricultural Engineering (AIIA)*, Springer International Publishing, 2020, pp. 793–801, https://doi.org/10.1007/978-3-030-39299-4_85.
- [27] S. Pearson, T.C. Camacho-Villa, R. Valluru, O. Gaju, M.C. Rai, I. Gould, E. Sklar, Robotics and autonomous systems for net zero agriculture, *Current Robotics Reports* 3 (2) (2022) 57–64, <https://doi.org/10.1007/s43154-022-00077-6>.
- [28] M. Gonzalez-de-Soto, L. Emmi, C. Benavides, I. García, P. Gonzalez-de-Santos, Reducing air pollution with hybrid-powered robotic tractors for precision agriculture, *Biosyst. Eng.* 143 (2016) 79–94, <https://doi.org/10.1016/j.biosystemseng.2016.01.008>.
- [29] Y. Tang, M. Chen, C. Wang, L. Luo, J. Li, G. Lian, X. Zou, Recognition and localization methods for vision-based fruit picking robots: a review, *Front. Plant Sci.* 11 (2020) 510, <https://doi.org/10.3389/fpls.2020.00510>.
- [30] S. Lyu, Q. Li, Z. Li, H. Liang, J. Chen, Y. Liu, H. Huang, Precision location-aware and intelligent scheduling system for monorail transporters in mountain orchards, *Agriculture* 13 (11) (2023) 2094, <https://doi.org/10.3390/agriculture13112094>.
- [31] H. Moreno, D. Andújar, Proximal sensing for geometric characterization of vines: a review of the latest advances, *Comput. Electron. Agric.* 210 (2023) 107901, <https://doi.org/10.1016/j.compag.2023.107901>.
- [32] A. Koirala, K.B. Walsh, Z. Wang, C. McCarthy, Deep learning for real-time fruit detection and orchard fruit load estimation: benchmarking of ‘MangoYOLO’, *Precis. Agric.* 20 (6) (2019) 1107–1135, <https://doi.org/10.1007/s11119-019-09642-0>.
- [33] H.K. Sharma, S. Gupta, M. Kapoor, Precision farming through IoT-enabled smart irrigation system for sustainable development, in: *Internet of Things*, Chapman and Hall/CRC, 2023, pp. 225–236, <https://doi.org/10.1201/9781003226888-17>.
- [34] O. Ponten, L. Xiao, J. Kutter, Y. Cui, C. Wählby, L. Behrendt, PACMan: a software package for automated single-cell chlorophyll fluorometry, *Cytometry* 105 (3) (2024) 203–213, <https://doi.org/10.1002/cyto.a.24808>.
- [35] N.R. Kolhalkar, V.L. Krishnan, R.G. Somkuwar, A.A. Pandit, Design of IoT-based mechatronic module for real-time yield monitoring during harvesting of greenhouse produce and vineyards, *Int. Rev. Mech. Eng.* 16 (2022) 48–55, <https://doi.org/10.15866/ireme.v16i1.21827>.
- [36] N. Shakoar, S. Lee, T.C. Mockler, High throughput phenotyping to accelerate crop breeding and monitoring of diseases in the field, *Curr. Opin. Plant Biol.* 38 (2017) 184–192, <https://doi.org/10.1016/j.pbi.2017.05.006>.
- [37] C. Xie, C. Yang, A review on plant high-throughput phenotyping traits using UAV-based sensors, *Comput. Electron. Agric.* 178 (2020) 105731, <https://doi.org/10.1016/j.compag.2020.105731>.

- [38] L. Liu, Y. Liu, X. He, W. Liu, Precision variable-rate spraying robot by using single 3D lidar in orchards, *Agronomy* 12 (10) (2022) 2509, <https://doi.org/10.3390/agronomy12102509>.
- [39] U. Bodkhe, S. Tanwar, P. Bhattacharya, N. Kumar, Blockchain for precision irrigation: opportunities and challenges, *Transactions on Emerging Telecommunications Technologies* 33 (10) (2022) e4059, <https://doi.org/10.1002/ett.4059>.
- [40] D.K. Singh, R. Sobti, Long-range real-time monitoring strategy for Precision Irrigation in urban and rural farming in society 5.0, *Comput. Ind. Eng.* 167 (2022) 107997, <https://doi.org/10.1016/j.cie.2022.107997>.
- [41] N.C. Eli-Chukwu, Applications of artificial intelligence in agriculture: a review, *Eng. Technol. Appl. Sci. Res.* 9 (4) (2019), <https://doi.org/10.48084/etasr.2756>.
- [42] C.R. Girelli, M. Hussain, D. Verweire, M.C. Oehl, J. Massana-Codina, M. S. Avendaño, F.P. Fanizzi, Agro-active endo-therapy treated *Xylella fastidiosa* subsp. pauca-infected olive trees assessed by the first 1H-NMR-based metabolomic study, *Sci. Rep.* 12 (1) (2022) 5973, <https://doi.org/10.1038/s41598-022-09687-8>.
- [43] L. Grandi, M. Oehl, T. Lombardi, V.R. De Michele, N. Schmitt, D. Verweire, D. Balmer, Innovations towards sustainable olive crop management: a new dawn by precision agriculture including endo-therapy, *Front. Plant Sci.* 14 (2023) 1180632, <https://doi.org/10.3389/fpls.2023.1180632>.
- [44] A.F. Lange, J. Peake, Precision agriculture, Position, Navigation, and Timing Technologies in the 21st Century: Integrated Satellite Navigation, Sensor Systems, and Civil Applications 2 (2020) 1735–1747, <https://doi.org/10.1002/9781119458555.ch56>.
- [45] R. Sinha, J.J. Quirós, S. Sankaran, L.R. Khot, High resolution aerial photogrammetry based 3D mapping of fruit crop canopies for precision inputs management, *Info. Process. Agric.* 9 (1) (2022) 11–23, <https://doi.org/10.1016/j.inpa.2021.01.006>.
- [46] J.C. Miranda, J. Gené-Mola, J. Arnó, E. Gregorio, AKFruitData: a dual software application for Azure Kinect cameras to acquire and extract informative data in yield tests performed in fruit orchard environments, *SoftwareX* 20 (2022) 101231, <https://doi.org/10.1016/j.softx.2022.101231>.
- [47] A.F. Jimenez Lopez, J.M. Salamanca, M.J. Quiroz Medina, O.E. Acevedo Pérez, Crops diagnosis using digital image processing and precision agriculture technologies, *Inge Cuc* 11 (1) (2015), <https://doi.org/10.17981/ingecuc.11.1.2015.06>.